

Integrating Rational Functions with Partial Fractions

- Factor the denominator first, then write one unknown per factor: a constant over each linear factor, an extra term over a repeated factor, and a linear numerator over an irreducible quadratic.
- The applied problems state their units. Carry them into your answer.
- Give exact values (ln, arctan, fractions) unless a decimal is requested, and add $+C$ to every indefinite answer.

1. Write $\frac{1}{(x-1)(x+3)}$ as a sum of two fractions, one over each linear factor.

Answer: _____

2. Evaluate $\int \frac{5}{(x-2)(x+3)} dx$.

Answer: _____

3. A reaction produces a compound at the rate $r(x) = \frac{6}{(x+1)(x+4)}$ grams per minute, where x is the number of minutes since the reaction started.

(a) Find the exact mass produced during the first 2 minutes.

Answer: _____ g

(b) Give that mass to three decimal places.

Answer: _____

4. Evaluate $\int \frac{x+4}{(x+2)^2} dx$. (The repeated factor needs two terms.)

Answer: _____

5. A population P , measured in thousands, satisfies a logistic model whose separated form requires $\int \frac{1}{P(10 - P)} dP$. Evaluate that integral, treating P as the variable.

Answer: _____

6. Evaluate $\int \frac{x^2 + 2x + 4}{x(x^2 + 4)} dx$. (The quadratic factor does not factor further, so its numerator is linear.)

Answer: _____

7. A filter removes a pollutant from a lake at the rate $r(t) = \frac{8}{(t+1)(t+3)}$ kilograms per year, where t is measured in years from installation.

(a) Find the exact mass removed during the first 3 years.

Answer: _____ kg

(b) Give that mass to three decimal places.

Answer: _____

8. A second filter removes a pollutant at the rate $r(x) = \frac{2}{x(x+2)}$ kilograms per year, for $x \geq 1$.

(a) Find the exact mass removed between $x = 1$ and $x = 3$.

Answer: _____

(b) For $b > 1$ the mass removed between $x = 1$ and $x = b$ equals $\ln\left(\frac{b}{b+2}\right) + \ln 3$. Find its limit as $b \rightarrow \infty$.

Answer: _____

(c) Explain what that limit says about the filter's lifetime total.

Answer: _____